



SDD LAB MID REPORT

SYED ARSHIYA TABASSUM
23ce02025
Group 2

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Q). The plan of a reinforced concrete **G+5 residential building** is shown in Figure. Perform

the Analysis and Design of the building using Staad.Pro Connect and provide the Detailing of reinforcement.

The following are some of the other pertinent details required for the analysis and design:

General details:

- Typical storey height: 3.3 m
- Depth of foundation: 2.5 m below Plinth level
- Height of plinth: 0.5 m
- Water table: 0.5 m from the foundation level
- Concrete grade for Super structure: M30
- Concrete grade for foundation: M25
- Main Reinforcing steel: Fe500
- Secondary Reinforcing steel: Fe415
- Unit weight of concrete: 25 kN/m³
- Unit weight of brick masonry: 20 kN/m³

Beam Column details:

- Beams
 - o Foundation to 2 storey: 0.3x0.6 m
 - o Storey 3 to storey 4: 0.3x0.45 m
 - o Storey 5 to top storey: 0.23x0.3 m
- Columns: 0.45x0.6 m
- Slab thickness: 0.18 m

Staircase details:

- Thickness of waist slab: 0.15 m
- Rise: 0.15 m
- Thread: 0.3 m

Loading details:

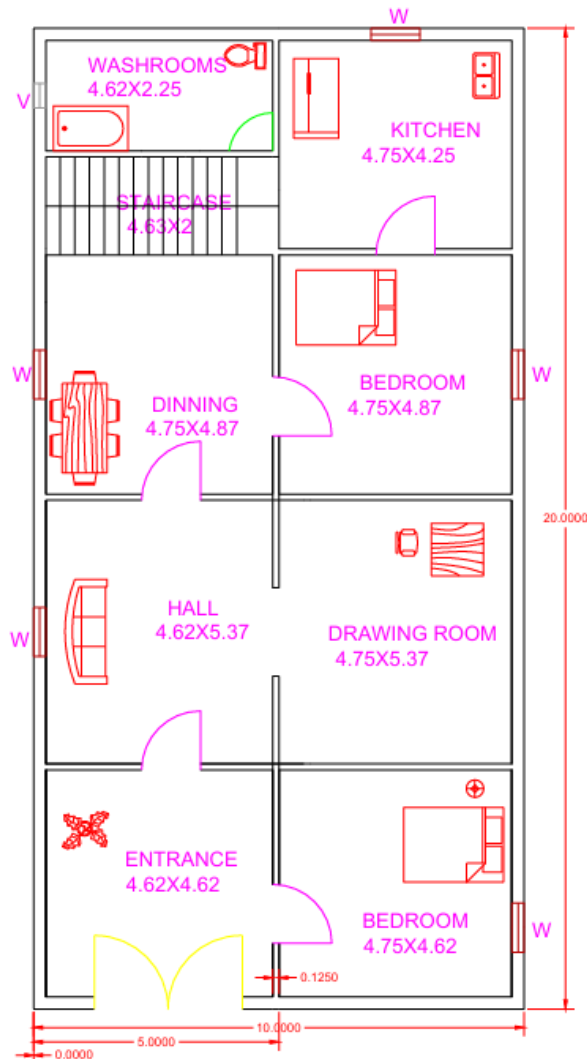
- Live load on floors: 3.0 kN/m²
- Live load on roof: 1.5 kN/m²
- Floor finish load: 1.5 kN/m²
- External wall thickness: 230 mm
- Internal wall thickness: 150 mm

For Wind and seismic details:

- Zone: Consider the hometown location
- Soil type: Medium soil

FUNCTIONAL PLAN:

PLAN LAYOUT



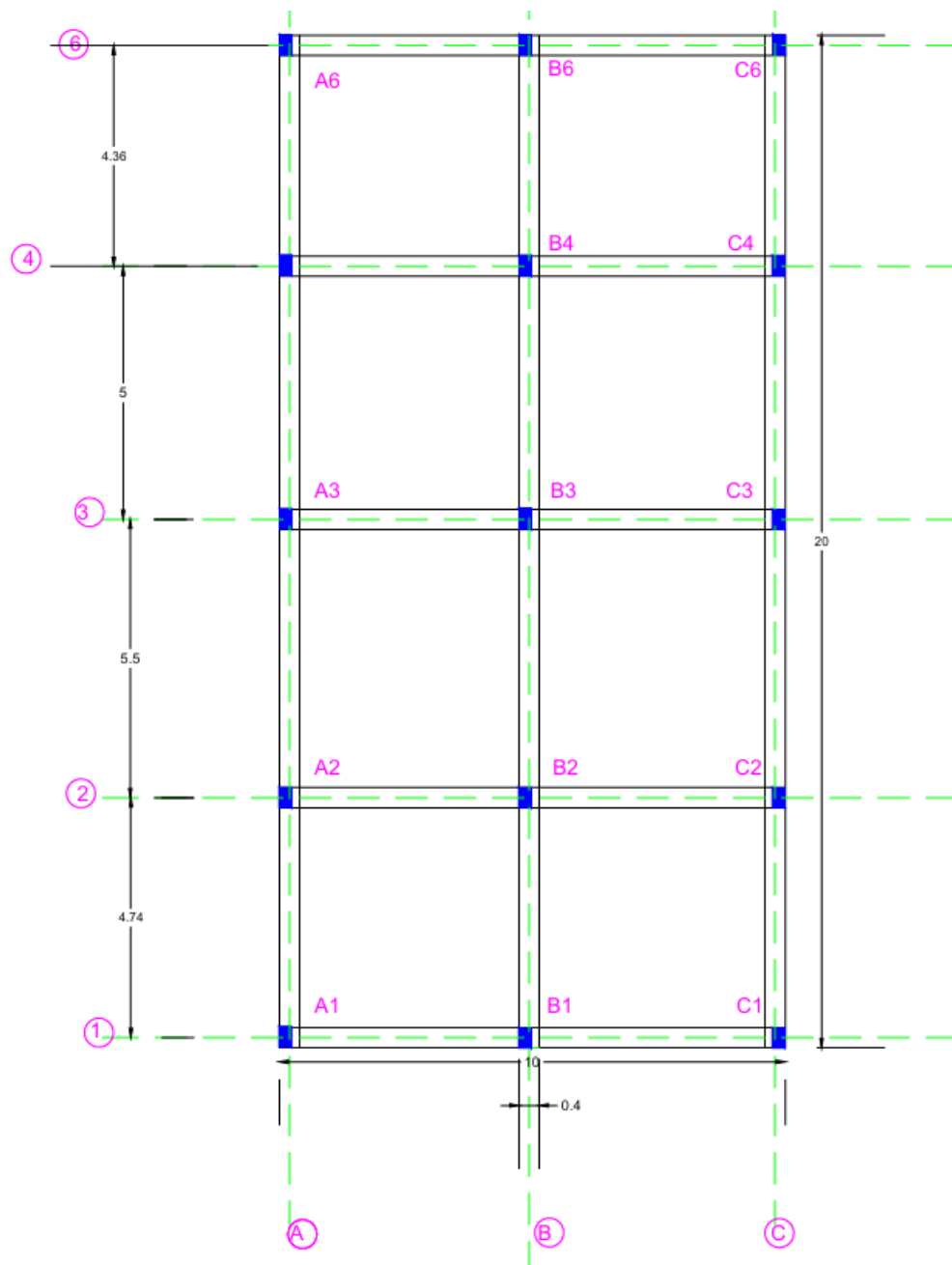
NOTES

All dimensions are in meters
Main door: 1.5X2.1
Room doors: 1.2X2.1
Toilet doors: 0.9X2.1
Windows: 1X1
Ventilators: 0.5X0.5

This is the functional plan for the residential building

The external walls are 230mm thick (full brick) and the internal walls are 150mm thick (half brick).

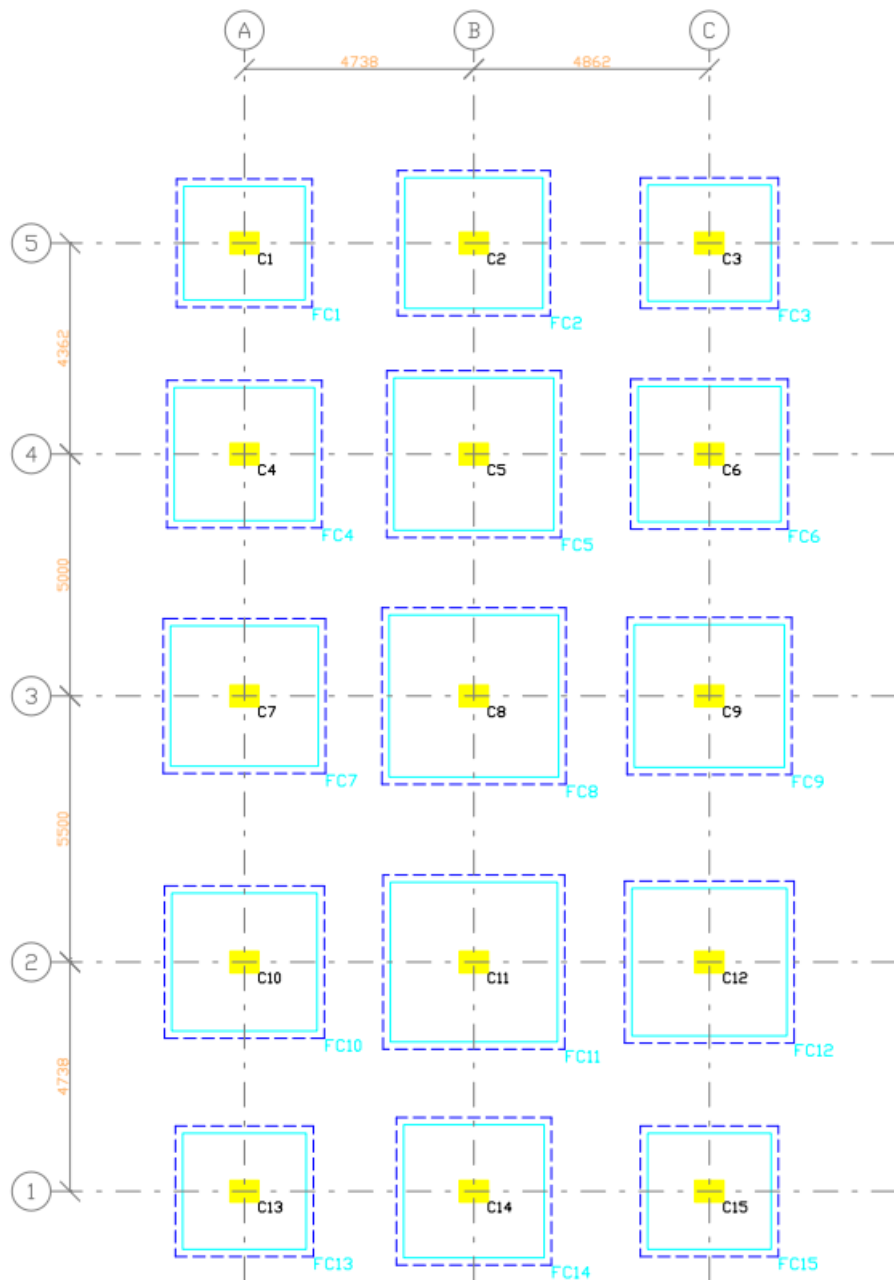
BEAM-COLUMN LAYOUT:



This is the BEAM-COLUMN grid layout

The grid spacing was ensured to be between 2 to 5.5 m for ensuring that the depth of the supporting beam provided is feasible and good enough to carry the applied load.

FOUNDATION LAYOUT PLAN:



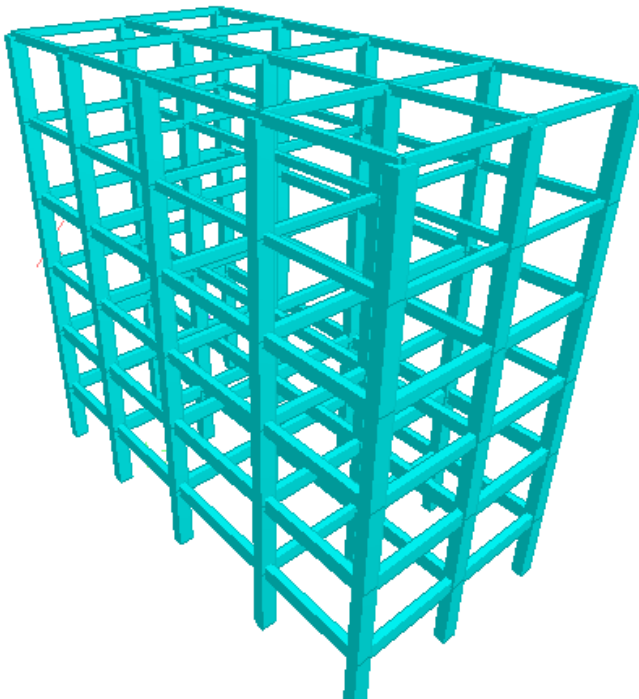
This is the layout plan of the foundation

All the foundations are provided as fixed supports.

The sectional properties of the foundations provided

Grp. No.	Footing Mark	Column Mark	Footing Type	Material Property	Column Size (mm)	Footing Size (LxBxD) (mm)
1	FC1	C1	Pad	M25 : Fe415	450 X 600	2500 x 2350 x 550
2	FC2	C2	Pad	M25 : Fe415	450 X 600	2850 x 2700 x 600
3	FC3	C3	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550
4	FC4	C4	Pad	M25 : Fe415	450 X 600	2900 x 2750 x 625
5	FC5	C5	Pad	M25 : Fe415	450 X 600	3300 x 3150 x 700
6	FC6	C6	Pad	M25 : Fe415	450 X 600	2950 x 2800 x 650
7	FC7	C7	Pad	M25 : Fe415	450 X 600	3050 x 2900 x 675
8	FC8	C8	Pad	M25 : Fe415	450 X 600	3500 x 3350 x 725
9	FC9	C9	Pad	M25 : Fe415	450 X 600	3100 x 2950 x 675
10	FC10	C10	Pad	M25 : Fe415	450 X 600	3000 x 2850 x 650
11	FC11	C11	Pad	M25 : Fe415	450 X 600	3450 x 3300 x 725
12	FC12	C12	Pad	M25 : Fe415	450 X 600	3200 x 3050 x 700
13	FC13	C13	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550
14	FC14	C14	Pad	M25 : Fe415	450 X 600	2900 x 2750 x 600
15	FC15	C15	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550

3-D MODEL OF THE STRUCTURE:



Calculations for loads on the structure:

1. Dead load:

- Self-weight of structural members:

For Beams:

- $0.3 \times 0.6 \text{ m} = 25 \times 0.3 \times 0.6 = 4.5 \text{ KN/m}$
- $0.3 \times 0.45 \text{ m} = 25 \times 0.3 \times 0.45 = 3.375 \text{ KN/m}$
- $0.23 \times 0.3 \text{ m} = 25 \times 0.23 \times 0.3 = 1.725 \text{ KN/m}$

For Column:

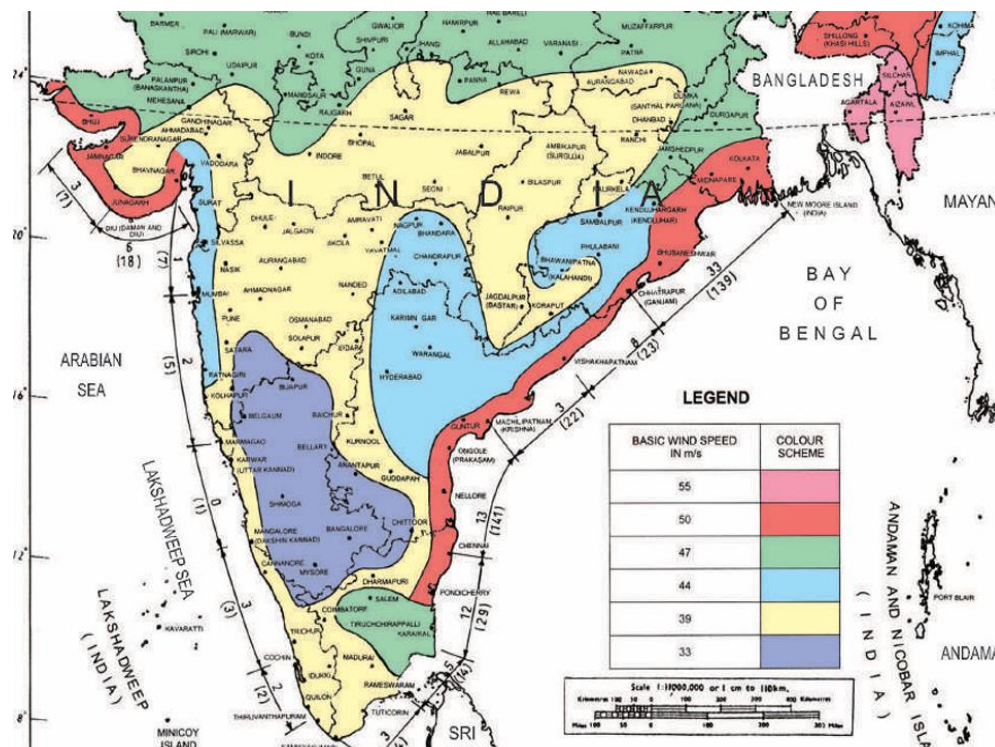
- $0.45 \times 0.6 \text{ m} = 25 \times 0.45 \times 0.6 = 6.75 \text{ KN}$
- Dead load of slab = $25 \times 0.18 = 4.5 \text{ KN/m}^2$
- Floor finish load = 1.5 KN/m^2
- Dead load of external wall = $0.23 \times 3.3 \times 20 = 15.18 \text{ KN/m}$
- Dead load of internal wall = $0.15 \times 3.3 \times 20 = 9.9 \text{ KN/m}$
- Staircase load = $0.15 \times 25 = 3.75 \text{ KN/m}^2$
- Total dead load = slab load + floor finish = $4.5 + 1.5 = 6 \text{ KN/m}^2$

2. Live load:

- Floor live load = 3 KN/m^2
- Roof live load = 1.5 KN/m^2

3. Wind load: (All design considerations are based on IS 875 (Part 3): 2015)

Basic Wind Speed (V_b) = 50 m/s (considering the structure located in Bhubaneswar)



Map for Basic wind speed in m/s (based on 50 Year return period) (from IS 875 part 3)

$$\text{Design wind speed } (V_z) = V_b \times K_1 \times K_2 \times K_3 \times K_4$$

V_z = Design wind speed at height z , in m/s;

K_1 = Probability factor;

K_2 = Terrain roughness and height factor;

K_3 = Topography factor

K_4 = Importance factor for the cyclonic region

The value of K_2 was found to be 1.099 by interpolating the values for the height of 16.5m from the table 2(**clause 6.3.2.2**).

Here, $K_1 = K_3 = K_4 = 1$.

Design wind speed (V_z) = $50 \times 1 \times 1.099 \times 1 \times 1 = 54.95$ m/s.

$$\text{Basic wind pressure } (P_z) = 0.6 (V_z)^2$$

$$P_z \text{ for } 10\text{m} = 0.6(V_z)^2$$

$$= 0.6(52.5)^2 \text{ (as } V_z = 50 \times 1 \times 1.05 \times 1 \times 1 = 52.5 \text{ m/s)}$$

$$= 1.653 \text{ KN/m}^2$$

$$P_z \text{ for } 16.5\text{m} = 0.6(V_z)^2$$

$$= 0.6(54.95)^2$$

$$= 1.811 \text{ KN/m}^2$$

Design wind pressure (P_d) = $K_d \times K_a \times K_c \times P_z \leq 0.7 P_z$

K_d = wind directionality factor

K_a = area averaging factor

K_c = combination factor

Let, $K_a = K_d = K_c = 1$

$P_d = P_z$.

Wind force (F) = $P_d \times (C_{pe} - C_{pi}) \times A$

C_{pe} = external pressure coefficient;

C_{pi} = internal pressure coefficient;

A = surface area of structural or cladding unit

P_d = design wind pressure

C_{pi} depends on the area of openings on the wall –

$C_{pi} = +0.2, -0.2$ when the openings are less than 5% of wall area

$C_{pi} = +0.5, -0.5$ when the openings are between 5%-20% of wall area

$C_{pi} = +0.7, -0.7$ when the openings are greater than 20% of wall area

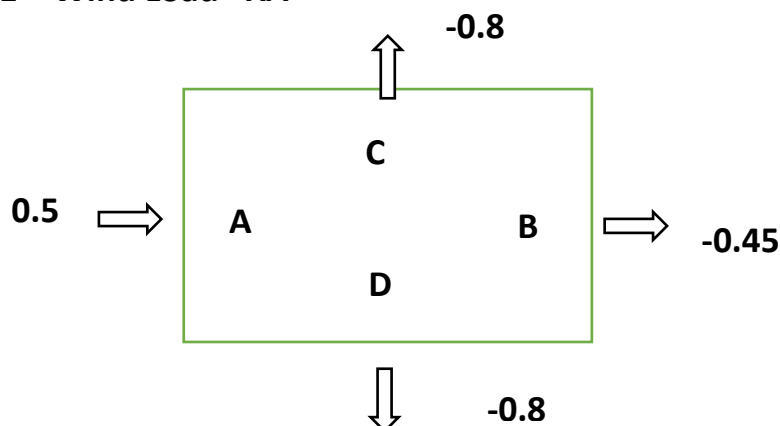
While calculating the wind loads on structure for each wind direction assumed two cases are considered- case A for internal pressure and case B for internal suction acting on the walls.

Opening area in all claddings < 5% of wall area. Hence, $C_{pi} = +0.2, -0.2$.

For external pressure coefficient calculation: $h = 16.5\text{m}$, $w = 19.6\text{m}$, $l = 9.6\text{m}$.

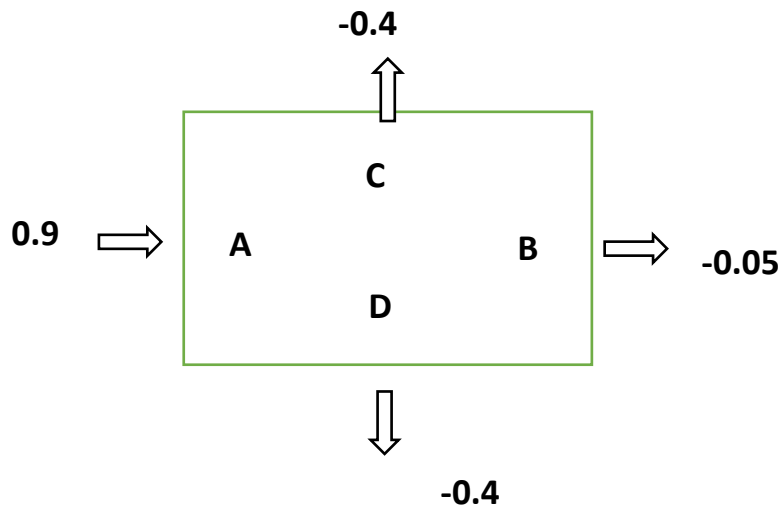
$h/w = 0.84$ and $l/w = 0.48$.

Case 1 – Wind Load +XA



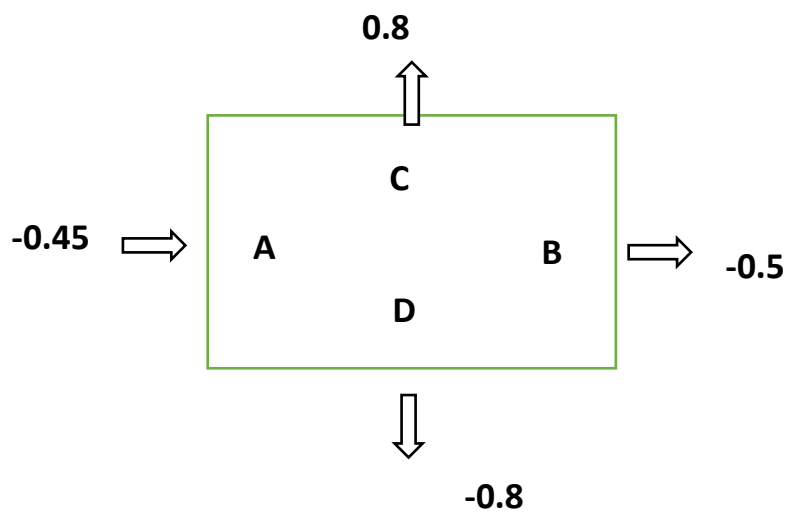
	A	B	C	D
C _{Pe}	+0.7	-0.25	-0.6	-0.6
C _{Pi}	0.2	0.2	0.2	0.2
C _{Pe} -C _{Pi}	0.5	-0.45	-0.8	-0.8

Case 2 – Wind Load +XB



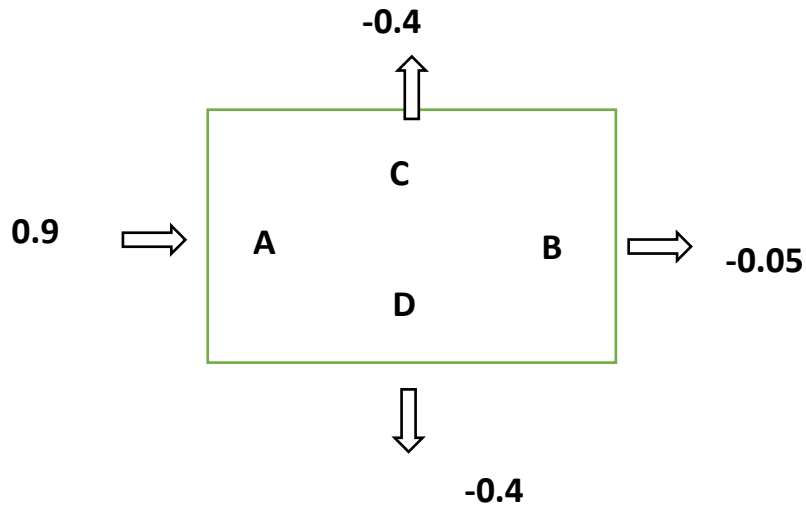
	A	B	C	D
C _{Pe}	+0.7	-0.25	-0.6	-0.6
C _{Pi}	-0.2	-0.2	-0.2	-0.2
C _{Pe} -C _{Pi}	0.9	-0.05	-0.4	-0.4

Case 3 – Wind Load -XA



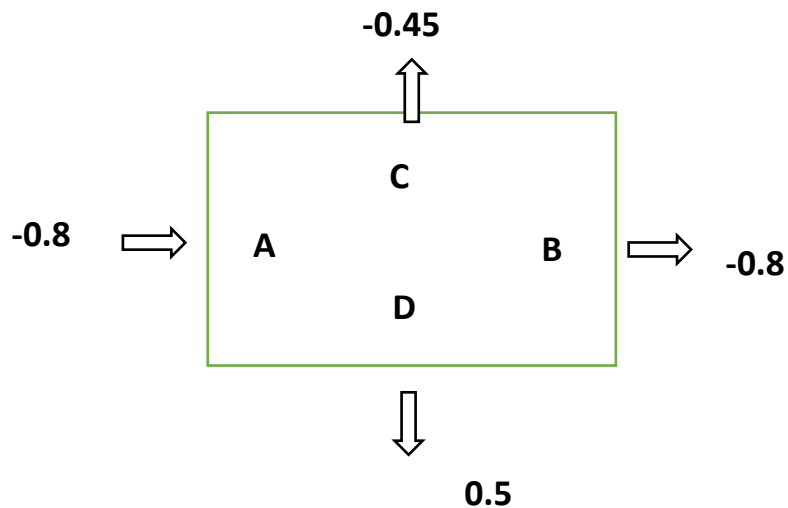
	A	B	C	D
C _{Pe}	-0.25	+0.7	-0.6	-0.6
C _{Pi}	0.2	0.2	0.2	0.2
C _{Pe} -C _{Pi}	-0.45	0.5	0.8	-0.8

Case 4 – Wind Load -XB



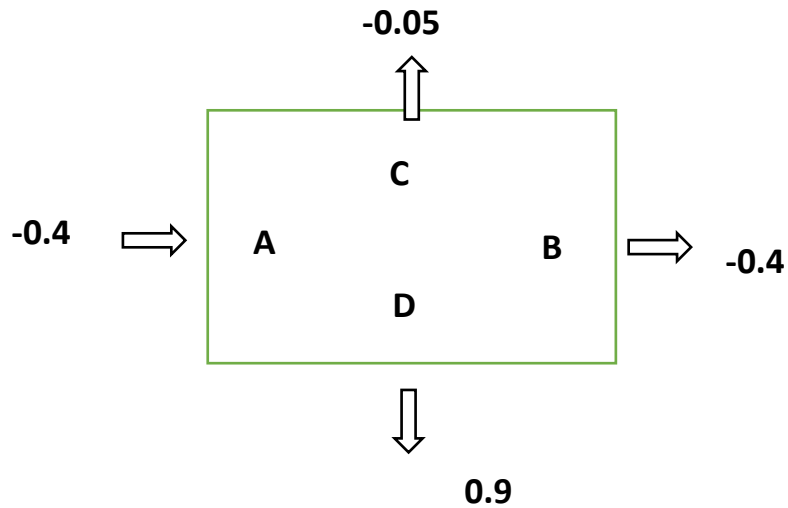
	A	B	C	D
C _{Pe}	+0.7	-0.25	-0.6	-0.6
C _{Pi}	- 0.2	-0.2	-0.2	-0.2
C _{Pe} -C _{Pi}	0.9	-0.05	-0.4	-0.4

Case 5 – Wind Load +ZA



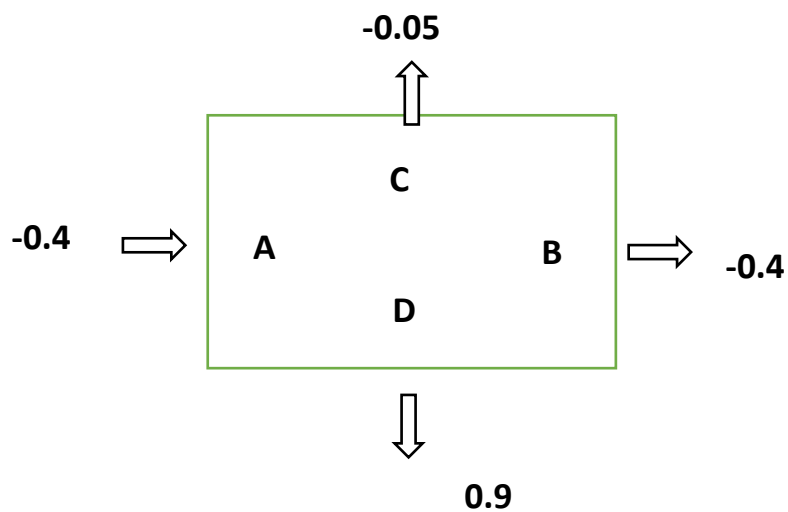
	A	B	C	D
C _{Pe}	-0.6	-0.6	-0.25	+0.7
C _{Pi}	0.2	0.2	0.2	0.2
C _{Pe} -C _{Pi}	-0.8	-0.8	-0.45	0.5

Case 6 – Wind Load +ZB



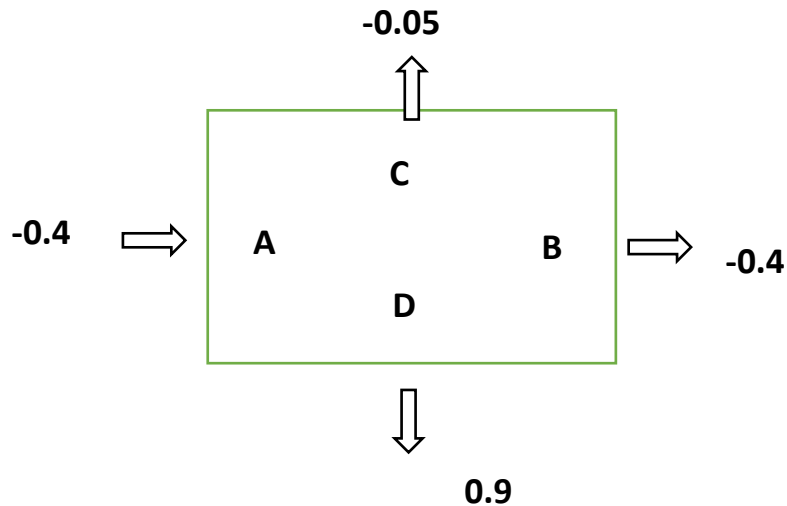
	A	B	C	D
C _{Pe}	-0.6	-0.6	-0.25	+0.7
C _{Pi}	-0.2	-0.2	- 0.2	- 0.2
C _{Pe} -C _{Pi}	-0.4	-0.4	-0.05	0.9

Case 7 – Wind Load -ZA



	A	B	C	D
C _{Pe}	-0.6	-0.6	+0.7	-0.25
C _{Pi}	0.2	0.2	0.2	0.2
C _{Pe} -C _{Pi}	-0.4	-0.4	-0.05	0.9

Case 8 – Wind Load -ZB



	A	B	C	D
C _{Pe}	-0.6	-0.6	+0.7	-0.25
C _{Pi}	-0.2	-0.2	- 0.2	-0.2
C _{Pe} -C _{Pi}	-0.4	-0.4	0.9	-0.05

4. Seismic load:

(All design considerations are based on IS 1893)

W_s (Seismic mass) = Dead Load + 0.25 Live Load on roof. (Eq 1)

Dead Load = Roof Load + 1 of Beam/Column/Wall Self Weight/Slab (Eq 2)

T = Fundamental Time Period.

F_s = Earthquake Load

$$F_s = A_n \times W_s$$

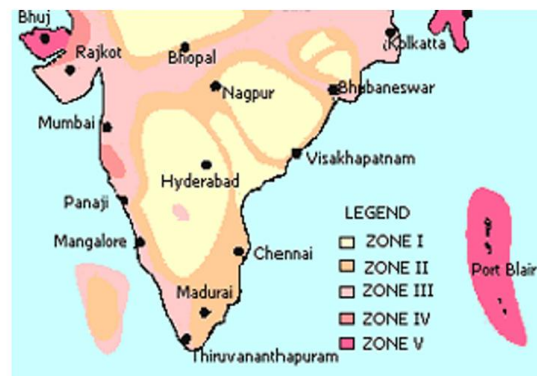
Due to Earthquake load a moment will be generated in the building which can be found by:

$$M_s = F_s \times e \quad (e = \text{Higher of the actual or minimum eccentricity})$$

Seismic zones in India:

- Zone II (mild zone)
- Zone III (moderate zone)
- Zone IV (severe zone)
- Zone V (very severe zone)

As the Building is in **Bhubaneswar** which happens to be in Zone III



$$Z=0.16$$

Seismic Weight at Roof level (W_s) = DL + LL

$$\frac{1}{2} \text{ of self-weight of 20 Columns (} W_{dc} \text{)} = 0.5 \times 20 \times 7.5 = 75 \text{ kN.}$$

$$\text{Similarly, Self-Weight of Beams (} W_{db} \text{)} = 14 \times 9.9 + 15 \times 15.18 = 366.3 \text{ kN.}$$

$$\text{Self-weight of roof slab (} W_{ds} \text{)} = 4.25 \text{ kN/m}^2 \times 15.18 \times 2 = 129.03 \text{ kN}$$

$$\text{Floor finish load at roof (} W_{dr} \text{)} = 1 \text{ kN/m}^2 \times 173.12 \text{ m}^2 = 173.12 \text{ kN}$$

$$\text{Load from parapet wall (} W_{dp} \text{)} = 4.75 \text{ kN/m} \times 2 \times (9.9 + 15.18) = 238.26 \text{ kN}$$

From Eq-1

$$DL = W_{dc} + W_{db} + W_{ds} + W_{dr} + W_{dp} = 981.71 \text{ kN}$$

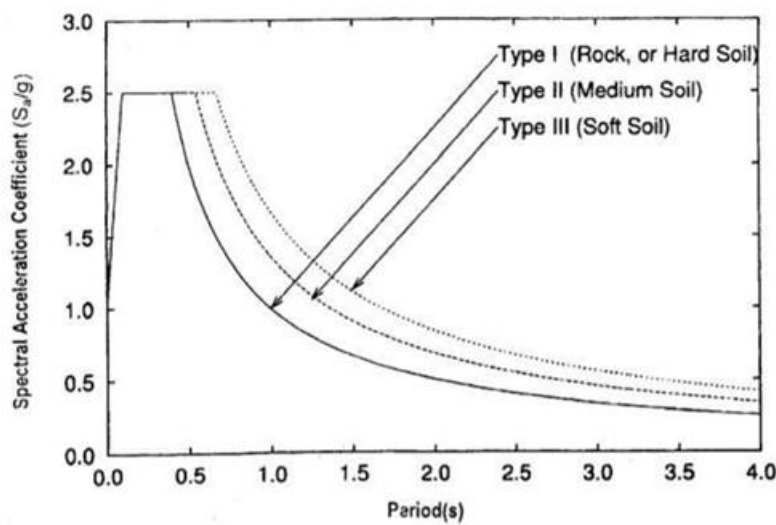
According to IS code we assume live load effect on seismic weight to be negligible, hence – seismic weight is equal to calculate dead load.

Seismic mass, $W_s = 981.71\text{kN}$

$$\text{Time period, } T_a = \frac{0.09h}{d^{1/2}}$$

$$T_{ax} = 0.09 * \frac{16.5}{9.61/2} = \mathbf{0.479s}$$

$$T_{az} = 0.09 * \frac{16.5}{19.61/2} = \mathbf{0.335s}$$



From Graph,

$$\frac{sa}{g} = 2.5 \text{ (for both X and Z directions)}$$

Building is Special RC Moment Resisting Frames (OMRF)

$I = 1.0$ (from Clause 6.4.2, for general buildings)

$Z = 0.16$ (from Table 3, for seismic zone III)

$R = 5$ (for RC buildings with special Moment resisting frames, from Table 9)

$$A_{nx} = \frac{ZIsa}{2gr} = 0.16 * 1 * \frac{2.5}{2*5} = 0.04$$

Base Shear:

$$V_{bx} = W_s \cdot A_{nx} = 981.71 * 0.04 = \mathbf{39.26\text{KN}}$$

Similarly,

$$V_{bz} = W_s \cdot A_{nz} = 981.71 * 0.04$$

$$= 39.26 \text{ kN}$$

Calculating Moments :

$$e_{min}(x) = 0.05 \cdot dx = 0.05 \cdot 9.9 \text{ m} = 0.495 \text{ m}$$

$$e_{min}(y) = 0.05 \cdot dy = 0.05 \cdot 15.18 \text{ m} = 0.759 \text{ m}$$

$$M_x = V_{bx} \cdot e_{min}(z) = 29.798 \text{ kNm}$$

$$M_z = V_{bz} \cdot e_{min}(x) = 19.433 \text{ kNm}$$

We assume lumped mass at the center of mass of the structure for analysis.

Seismic loads on the structure are applied at the center of stiffness of the structure. (here the structure is symmetric therefore the center of mass and center of stiffness coincide therefore, $e_{static} = 0$)

Position of Master Node for application of the seismic loads on the structure is centroid, i.e. (6.617 , 3, 7.061)

General conditions followed for design to find the seismic responses (for economical and safe design)

- In frequent earthquakes (small magnitude) – accept no damage
- In moderate magnitude earthquakes – accept minor damage
- In large magnitude earthquake (very high return period) – accept major damage but no collapse

Seismic Load Combination:

For Irregular Structure:

- $\pm EL_x \pm 0.3EL_y \pm 0.3EL_z$
- $\pm EL_y \pm 0.3EL_x \pm 0.3EL_z$
- $\pm EL_z \pm 0.3EL_x \pm 0.3EL_y$

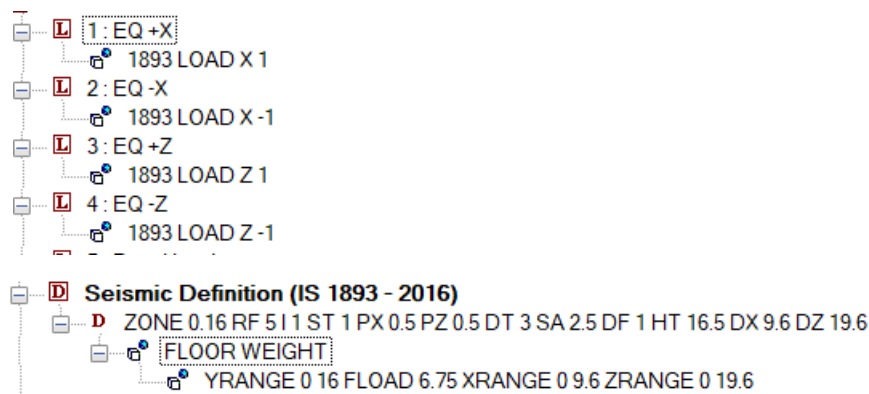
For regular buildings:

- $1.5DL + 1.5[F_{sx} + M_s]$
- $1.5DL + 1.5[F_{sx} - M_s]$
- $1.5DL + 1.5[-F_{sx} + M_s]$

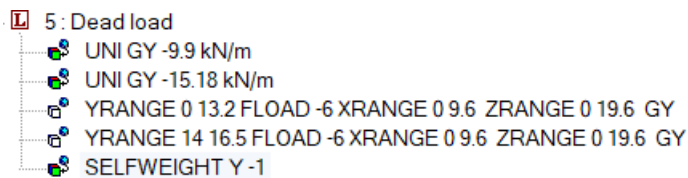
- $1.5DL + 1.5[-F_{sx} - M_s]$
- $1.5DL + 1.5[F_{sz} + M_s]$
- $1.5DL + 1.5[F_{sz} - M_s]$
- $1.5DL + 1.5[-F_{sz} + M_s]$
- $1.5DL + 1.5[-F_{sz} - M_s]$

LOADING CASES DETAILS:

1-4: Earthquake Load



5: Dead Load (DL)

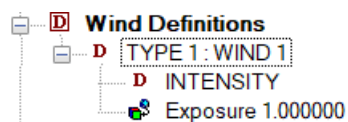


6: Live Load (LL)

6: Live load
 YRANGE 0 16.5 FLOAD -3 X RANGE 0 9.6 Z RANGE 0 19.6 GY

7: Wind Load (WL)

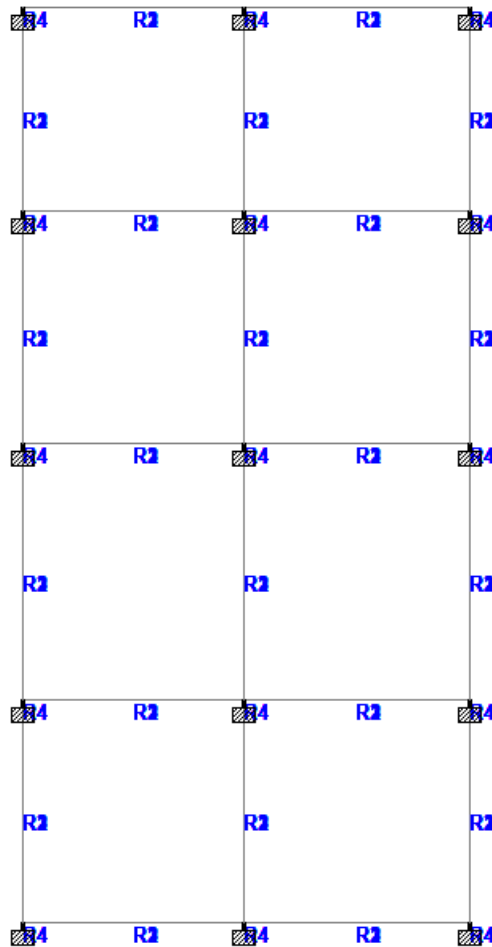
7: Wind load
 WIND LOAD X 1.4 TYPE 1 XR 0 9.6 YR 0 16.5 ZR 0 19.6



8: Load combinations including Earthquake loads

		8 : COMB - 1.5 Dead + 1.5 Live
		9 : COMB - 1.2 Dead + 1.2 Live + 1.2 Wind
		10 : COMB - 1.2 Dead + 1.2 Live + -1.2 Wind
		11 : COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (1)
		12 : COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (2)
		13 : COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (3)
		14 : COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (4)
		15 : COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (1)
		16 : COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (2)
		17 : COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (3)
		18 : COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (4)
		19 : COMB - 1.5 Dead + 1.5 Wind
		20 : COMB - 1.5 Dead + -1.5 Wind
		21 : COMB - 1.5 Dead + 1.5 Seismic-H (1)
		22 : COMB - 1.5 Dead + 1.5 Seismic-H (2)
		23 : COMB - 1.5 Dead + 1.5 Seismic-H (3)
		24 : COMB - 1.5 Dead + 1.5 Seismic-H (4)
		25 : COMB - 1.5 Dead + -1.5 Seismic-H (1)
		26 : COMB - 1.5 Dead + -1.5 Seismic-H (2)
		27 : COMB - 1.5 Dead + -1.5 Seismic-H (3)
		28 : COMB - 1.5 Dead + -1.5 Seismic-H (4)
		18 : COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (4)
		19 : COMB - 1.5 Dead + 1.5 Wind
		20 : COMB - 1.5 Dead + -1.5 Wind
		21 : COMB - 1.5 Dead + 1.5 Seismic-H (1)
		22 : COMB - 1.5 Dead + 1.5 Seismic-H (2)
		23 : COMB - 1.5 Dead + 1.5 Seismic-H (3)
		24 : COMB - 1.5 Dead + 1.5 Seismic-H (4)
		25 : COMB - 1.5 Dead + -1.5 Seismic-H (1)
		26 : COMB - 1.5 Dead + -1.5 Seismic-H (2)
		27 : COMB - 1.5 Dead + -1.5 Seismic-H (3)
		28 : COMB - 1.5 Dead + -1.5 Seismic-H (4)
		29 : COMB - 0.9 Dead + 1.5 Wind
		30 : COMB - 0.9 Dead + -1.5 Wind
		31 : COMB - 0.9 Dead + 1.5 Seismic-H (1)
		32 : COMB - 0.9 Dead + 1.5 Seismic-H (2)
		33 : COMB - 0.9 Dead + 1.5 Seismic-H (3)
		34 : COMB - 0.9 Dead + 1.5 Seismic-H (4)
		35 : COMB - 0.9 Dead + -1.5 Seismic-H (1)
		36 : COMB - 0.9 Dead + -1.5 Seismic-H (2)
		37 : COMB - 0.9 Dead + -1.5 Seismic-H (3)
		38 : COMB - 0.9 Dead + -1.5 Seismic-H (4)

BEAM DESIGN



Taking beam no.206

Beam no. = 206. Section: Rect 0.30x0.23

Length = 4.863m

Section diagrams are approximate representation and not to the scale.

Physical Properties

Ax(cm ²)	690.000	Ix(cm ⁴)	64595.305
Ay(cm ²)	575.000	Iy(cm ⁴)	30417.504
Az(cm ²)	575.000	Iz(cm ⁴)	51750.004
D(cm)	30.000	W(cm)	23.000

Assign/Change Property

Material Properties

Elasticity(kN/mm ²)	21.719	Density(kg/m ³)	2402.615
Poisson	0.17	Alpha	1e-05

CONCRETE

Assign Material

Density of Concrete = 2550kg/m³

Self-weight of concrete = 2550 x 9.81 = 25kN/m³

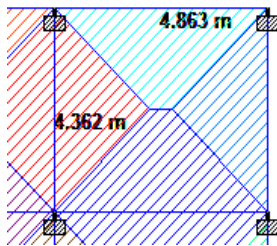
UDL due to self-weight = $25 \times 0.3 \times 0.23 = 1.725 \text{ kN/m}$

Live load on slab = 1.5 kN/m^2

Pressure due to slab = $0.18 \times 25 = 4.5 \text{ kN/m}^2$

Dead load due to floor finish = 1 kN/m^2

$w = 7 \text{ kN/m}^2$ (total pressure to be distributed on the slab and transferred to the beams)



Load transferred to beam no 206 from slab

Load transferred to beam no 206 from slab

$$\beta = L_y/L_x = 4.863 / 3 = 1.621$$

$$\text{Trapezoidal load} = (wL_x/2) * (1 - 1/2\beta)$$

$$= (7 * 3/2) * (1 - 1/3.242)$$

$$= 7.261 \text{ kN/m}$$

Check for shear:

$$R_1 + R_2 = (1.725 \times 4.863) + 1/2 \times 4.863 \times 7.261$$

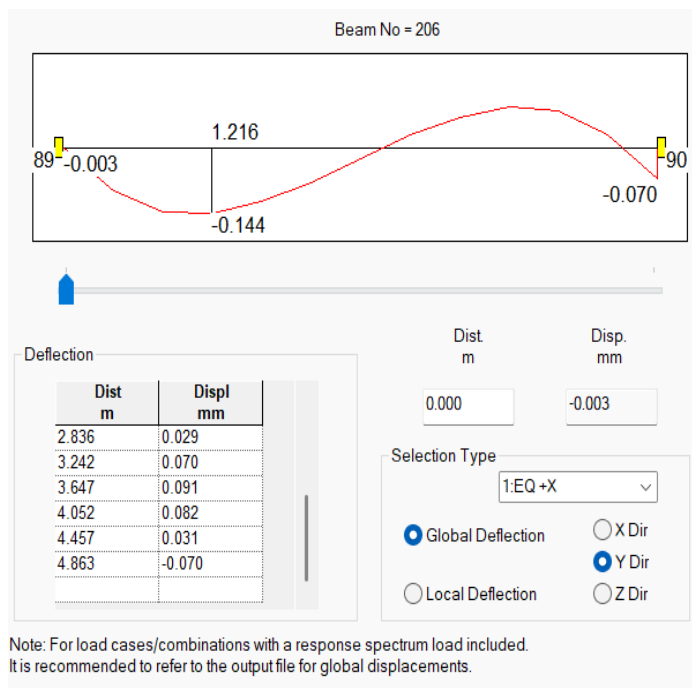
$$R_1 + R_2 = 26.043 \text{ kN}$$

$$R_1 = R_2 \text{ (Assuming simply supported beam)}$$

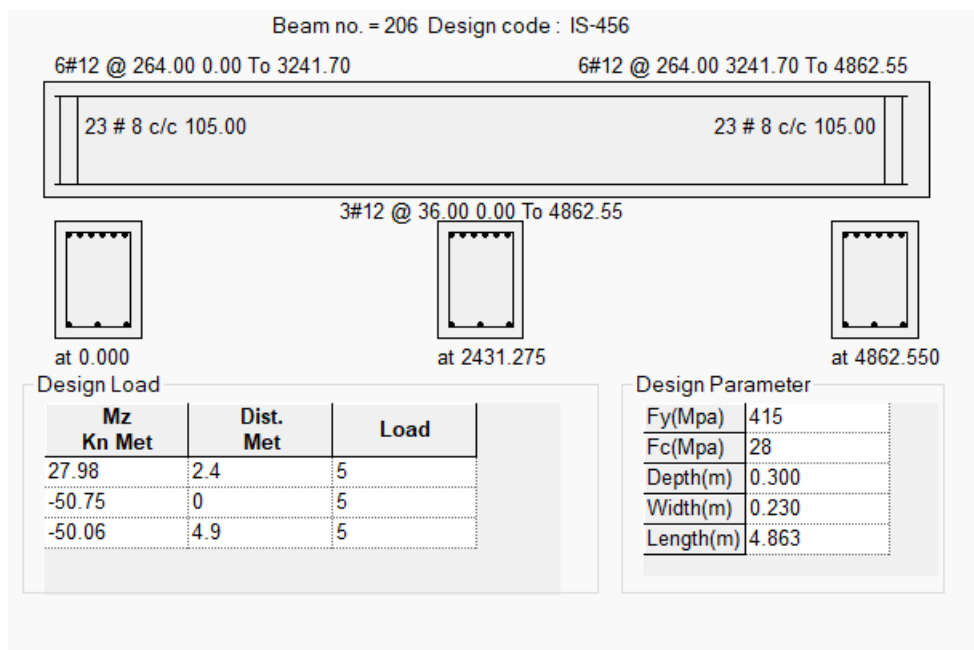
$$\text{So, } R_1 = R_2 = 13.021 \text{ kN}$$

The actual values obtained from StaadPro are 20.1kN, which are very close to the theoretical value found by hand calculation.

Maximum Global Deflection in the beam is 0.144mm



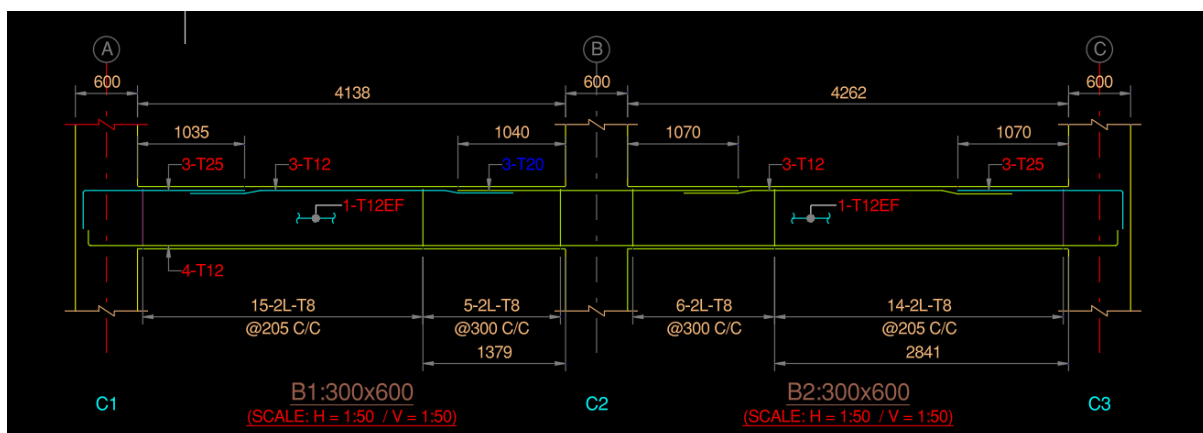
Concrete Design as obtained from STAAD Pro:



Summary for Beam No. 206 from STAAD Pro

sdd 8 march assignment new - Beam End Forces:									
Summary / Envelope /									
Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kN-m	My kN-m	Mz kN-m	
206	5 Dead load	86	27.896	74.395	-0.005	-0.009	0.011	66.079	
		89	-27.896	72.558	0.005	0.009	0.011	-61.728	
	6 Live load	86	7.539	16.971	-0.003	-0.004	0.006	16.793	
		89	-7.539	16.696	0.003	0.004	0.007	-16.141	
	7 Wind load	86	-0.060	-0.002	-2.389	-0.222	5.580	-0.005	
		89	0.060	0.002	2.389	0.222	5.738	-0.005	
	1 EQ +X	89	-0.472	-1.682	0.445	0.003	-0.972	-4.038	
		90	0.472	1.682	-0.445	-0.003	-1.190	-4.141	
	2 EQ -X	89	0.472	1.682	-0.445	-0.003	0.972	4.038	
		90	-0.472	-1.682	0.445	0.003	1.190	4.141	
	3 EQ +Z	89	-0.298	-0.004	0.457	-0.034	-1.267	-0.010	
		90	0.298	0.004	-0.457	0.034	-0.955	-0.010	
	4 EQ -Z	89	0.298	0.004	-0.457	0.034	1.267	0.010	
		90	-0.298	-0.004	0.457	-0.034	0.955	0.010	
	5 Dead load	89	22.495	58.722	-0.028	-0.204	0.068	50.749	
		90	-22.495	58.440	0.028	0.204	0.066	-50.064	
	6 Live load	89	4.326	8.769	-0.016	-0.092	0.041	8.646	
		90	-4.326	8.952	0.016	0.092	0.038	-9.090	
	7 Wind load	89	2.041	-4.902	1.128	0.008	-2.468	-11.770	
		90	-2.041	4.902	-1.128	-0.008	-3.017	-12.067	

Reinforcement Detailing:



COLUMN DESIGN:

Reinforcement obtained for all columns from staad pro is as follows-

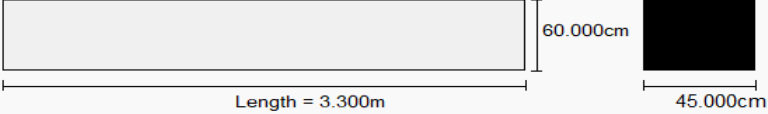
- Main rebar 4-20mm diameter
- Lateral ties 8mm diameter @ 250mm C-C spacing

Taking column No. 185

Cross Section = 0.6m x 0.45m

Length= 3.3m (M30)

Beam no. = 185. Section: Rect 0.60x0.45



Length = 3.300m

Section diagrams are approximate representation and not to the scale.

Physical Properties

Ax(cm ²)	2700.000	Ix(cm ⁴)	984074.375
Ay(cm ²)	2250.000	Iy(cm ⁴)	455624.969
Az(cm ²)	2250.000	Iz(cm ⁴)	810000.125
D(cm)	60.000	W(cm)	45.000

Assign/Change Property

Material Properties

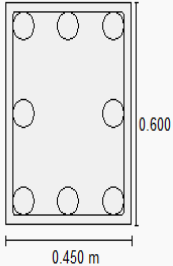
Elasticity(kN/mm ²)	21.719	Density(kg/m ³)	2402.615
Poisson	0.17	Alpha	1e-05

CONCRETE

Assign Material

Concrete design obtained from STAAD:

Beam no. = 185 Design code : IS-456



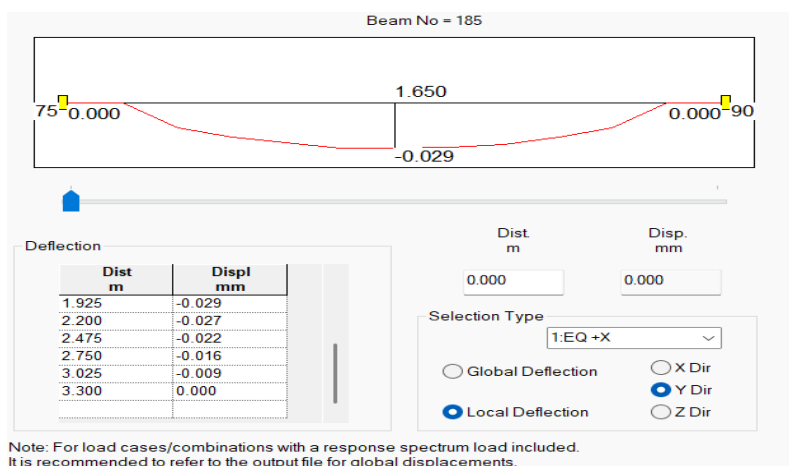
Design Load

Load	5
Location	End 2
Pu(Kns)	114.48
Mz(Kns-Mt)	50.18
My(Kns-Mt)	46.23

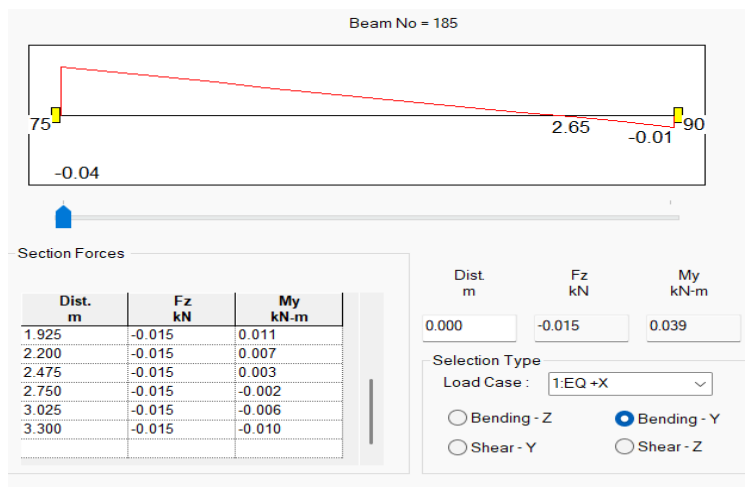
Design Parameter

Fy(Mpa)	415
Fc(Mpa)	28
As Reqd(mm ²)	856
As (%)	0.335
Bar Size	12
Bar No	8

Maximum Local deflection of column 185 along Y direction:



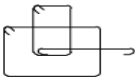
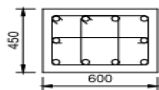
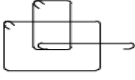
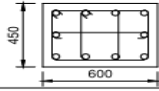
Bending of the column along Y direction:



Summary for the Column 185 obtained from STAAD pro

Beam	L/C	Dist m	x mm	y mm	z mm	Resultant mm
185	1 EQ +X	3.300	0.000	0.000	0.000	0.000
		0.000	0.000	0.000	0.000	0.000
		0.825	-0.001	-0.022	0.000	0.022
		1.650	-0.000	-0.029	-0.001	0.029
	2 EQ -X	2.475	-0.001	-0.022	-0.000	0.022
		3.300	0.000	0.000	0.000	0.000
		0.000	0.000	0.000	0.000	0.000
		0.825	0.001	0.022	-0.000	0.022
	3 EQ +Z	1.650	-0.000	0.029	0.001	0.029
		2.475	0.001	0.022	0.000	0.022
		3.300	0.000	0.000	0.000	0.000
		0.000	0.000	0.000	0.000	0.000
	4 EQ -Z	0.825	-0.001	0.000	0.039	0.039
		1.650	-0.000	0.000	0.050	0.050
		2.475	-0.001	0.000	0.038	0.038
		3.300	0.000	0.000	0.000	0.000
	5 Dead load	0.000	0.000	0.000	0.000	0.000
		0.825	0.000	-0.040	0.042	0.058
		1.650	0.001	-0.102	0.140	0.173
		2.475	0.001	-0.112	0.167	0.201
	6 Live load	3.300	0.000	0.000	0.000	0.000
		0.000	0.000	0.000	0.000	0.000
		0.825	-0.000	-0.004	0.007	0.009
		1.650	-0.000	-0.016	0.026	0.030
		2.475	-0.000	-0.019	0.030	0.035

Column detailed drawing:

0 M TO 3.3 M	M30 : Fe500 (M) : Fe415 (S) , COVER = 50MM 
	LINKS T8 @ 175
	 10-T12
-3 M TO 0 M	M30 : Fe500 (M) : Fe415 (S) , COVER = 50MM 
	LINKS T8 @ 175
	 10-T12
COLUMN MARKED	C1

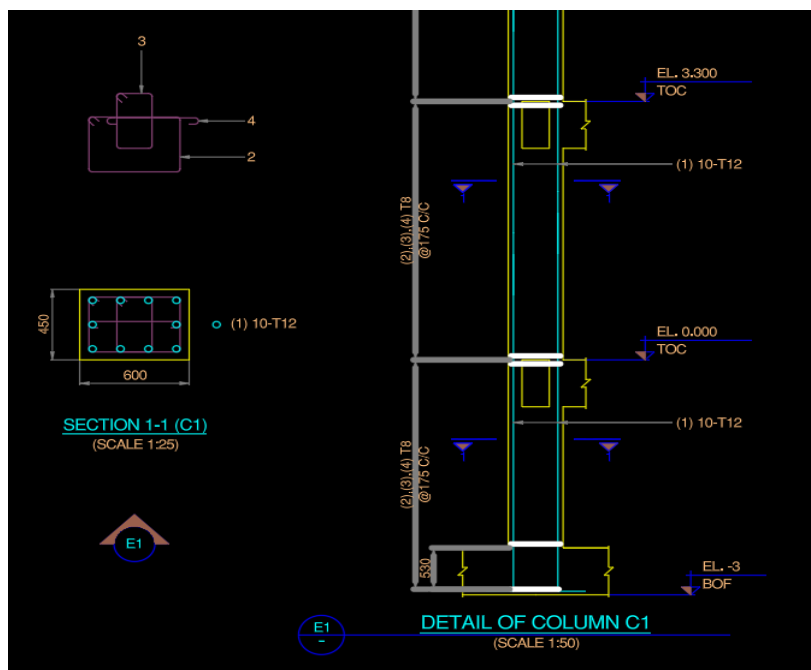
COLUMN AND WALL SCHEDULE

(SCALE 1:25)

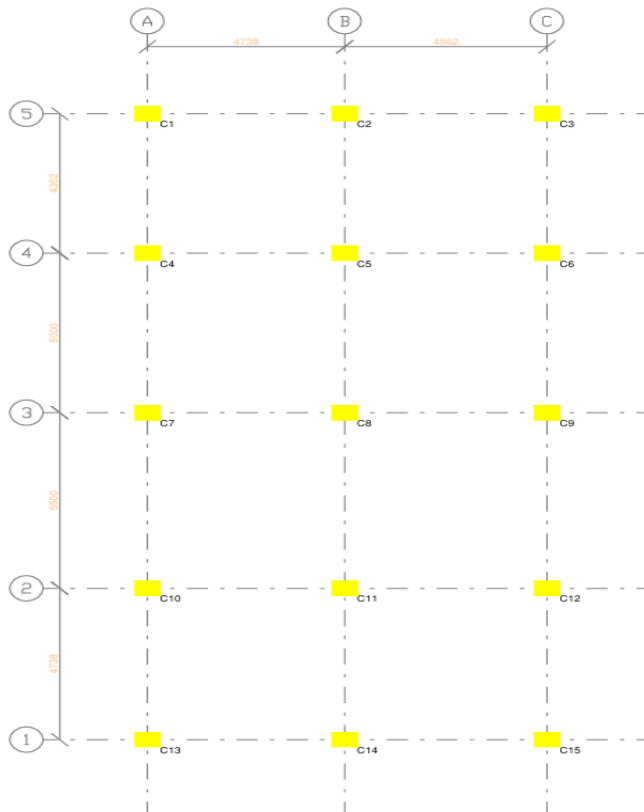
NOTES:

1. (M) – STEEL GRADE FOR MAIN REINFORCEMENT
2. (S) – STEEL GRADE FOR SHEAR REINFORCEMENT/LINKS

Column elevation C1 -3 to 3m.



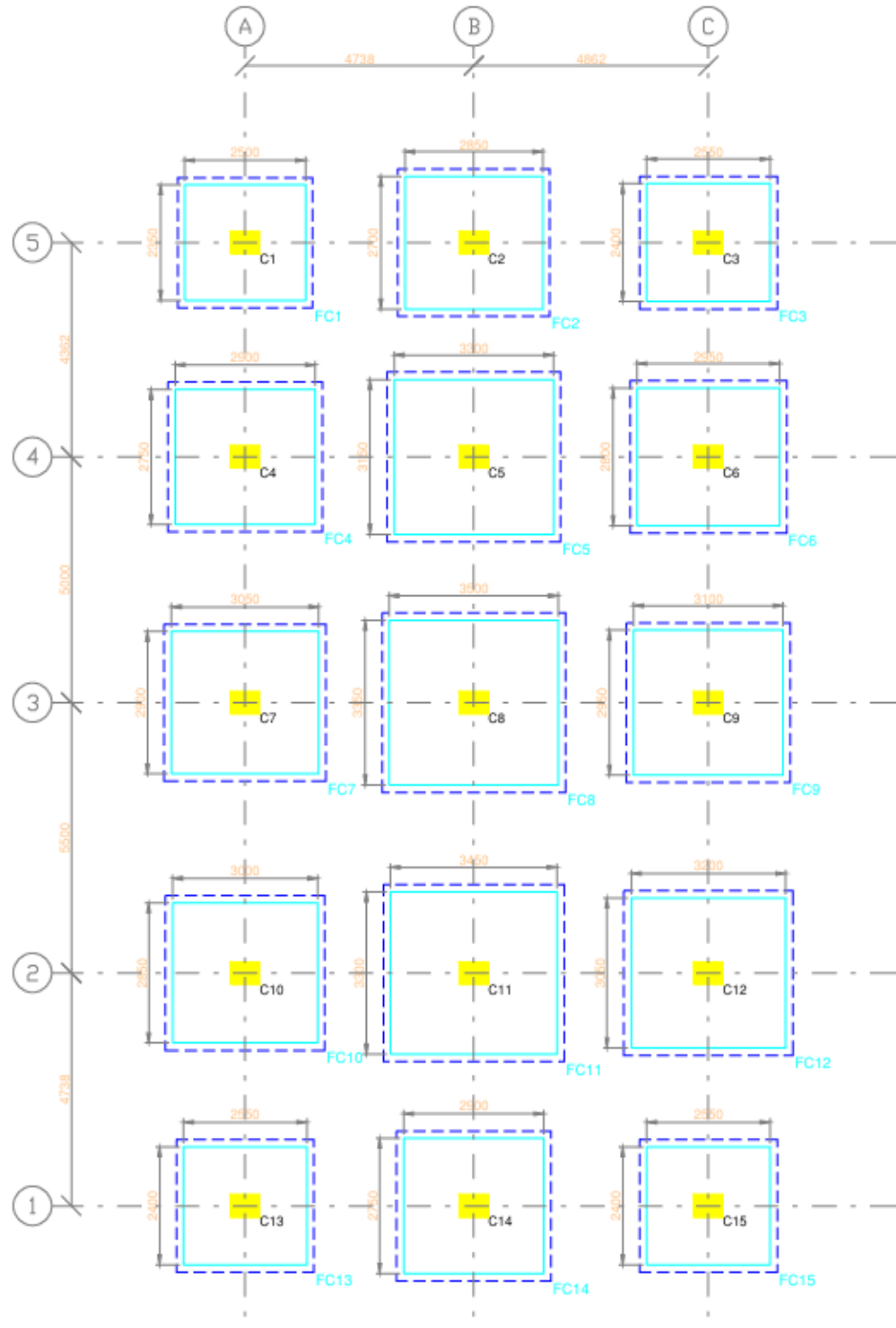
Column Layout:



FOUNDATION DESIGN:

- The concrete used in the footing is M30 and the grade of steel used is Fe500.
- The SBC of the soil is assumed to be 120 kPa.
- Depth of soil above footing is 500 mm.
- Coefficient of friction = 0.5
- Factor of safety against sliding and overturning = 1.5

Detailed Drawing:



Grp. No.	Footing Mark	Column Mark	Footing Type	Material Property	Column Size (mm)	Footing Size (LxBxD) (mm)
1	FC1	C1	Pad	M25 : Fe415	450 X 600	2500 x 2350 x 550
2	FC2	C2	Pad	M25 : Fe415	450 X 600	2850 x 2700 x 600
3	FC3	C3	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550
4	FC4	C4	Pad	M25 : Fe415	450 X 600	2900 x 2750 x 625
5	FC5	C5	Pad	M25 : Fe415	450 X 600	3300 x 3150 x 700
6	FC6	C6	Pad	M25 : Fe415	450 X 600	2950 x 2800 x 650
7	FC7	C7	Pad	M25 : Fe415	450 X 600	3050 x 2900 x 675
8	FC8	C8	Pad	M25 : Fe415	450 X 600	3500 x 3350 x 725
9	FC9	C9	Pad	M25 : Fe415	450 X 600	3100 x 2950 x 675
10	FC10	C10	Pad	M25 : Fe415	450 X 600	3000 x 2850 x 650
11	FC11	C11	Pad	M25 : Fe415	450 X 600	3450 x 3300 x 725
12	FC12	C12	Pad	M25 : Fe415	450 X 600	3200 x 3050 x 700
13	FC13	C13	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550
14	FC14	C14	Pad	M25 : Fe415	450 X 600	2900 x 2750 x 600
15	FC15	C15	Pad	M25 : Fe415	450 X 600	2550 x 2400 x 550

Footing design summary:

Footing No: FC1
 Column No: C1 (450 mm X 600 mm)
 Footing Type: Pad
 Concrete Grade: M25
 Steel Grade: Fe415

Size (LxBxD)	Bot @ L	Bot @ B	Top @ L	Top @ B	Shear @ L	Shear @ B	SFR
2500x2350x550	25-T10 @ 100	23-T10 @ 115	10-T10 @ 265	11-T10 @ 250	-	-	-

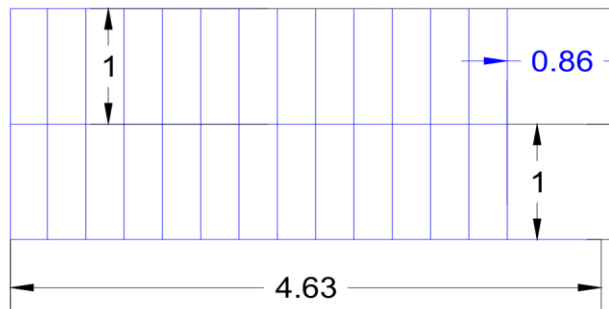
Footing No: FC2
 Column No: C2 (450 mm X 600 mm)
 Footing Type: Pad
 Concrete Grade: M25
 Steel Grade: Fe415

Size (LxBxD)	Bot @ L	Bot @ B	Top @ L	Top @ B	Shear @ L	Shear @ B	SFR
2850x2700x600	23-T12 @ 125	24-T12 @ 125	13-T10 @ 225	14-T10 @ 220	-	-	-

STAIRCASE DESIGN (DOG LEGGED TYPE):

Assumptions

- $f_{ck} = 20 \text{ MPa}$, $f_y = 415 \text{ MPa}$
- Total width = 2m
- Riser = 150 mm
- Tread width = 300 mm
- Landing = 860 mm
- Supports are simply supported
- Live load = 3 kN/m²



Geometric Design

- $(T^2 + R^2)^{1/2} = 335.41\text{mm}$
- Number of steps = (total rise/rise) = $2100/150 = 14$
- Effective span length = 4630mm
Assuming $l/d = 20$ (for slab)
 $d_{\text{eff}} \geq 4630/20 = 231.5\text{ mm}$
Considering, clear cover = 25 mm, main rebar diameter = 12mm, total depth = 300mm
So, effective depth = total depth (D) – nominal cover – rebar dia./2
 $= (300 - 25 - 12/2)\text{ mm}$
 $= 269\text{mm} \geq 231.5\text{mm}$. Hence, ok.
So, $d = 269\text{mm}$, $D = 300\text{mm}$.

Loads on going portion:

- Self-weight of Waist slab = $25 \times D \times (T^2 + R^2)^{1/2} / T$
 $= 25 \times 0.3 \times 0.335/0.15 = 16.75\text{kN/m}^2$
- Weight of Steps = $25 \times 0.5 \times (R \times T) / T$
 $= 25 \times 0.5 \times 0.15 = 1.875\text{ kN/m}^2$
- Load due to floor finishes:
Floor finish = 0.8 kN/m^2
- Live load = 3 kN/m^2

DESIGN LOAD = $1.5 \times (\text{total dead load} + \text{total live load})$

$$= 1.5 \times (16.75 + 1.875 + 0.8 + 3) = 33.6375\text{ kN/m}^2$$

Loads on Landings

- Dead load = $25 \times 0.3 = 7.5\text{ kN/m}^2$
- Floor Finish = 0.8 kN/m^2

- Live load = 3 kN/m²

DESIGN LOAD ON LANDING = 1.5 (DL + FF + LL)

$$= 1.5 \times (7.5 + 0.8 + 3) = 16.95 \text{ kN/m}^2$$

Moments and Reinforcement calculations:

- Total load on each flight = $(16.95 \times 1.193) \times 2 + (33.63 \times 2.79) = 134.27 \text{ kN}$
- Reactions at supports: $R_1 + R_2 = 134.27 \text{ kN}$
- As loading is symmetrical, $R_1 = R_2 = 67.135 \text{ kN}$
- UDL : $W_1 = 33.63 \text{ kN/m}$
 $W_2 = 16.95 \text{ kN/m}$
- Maximum bending moment in beam will be at center of span,

$$M_d = R_2 \times (2.79/2 + 1.193) - W_2 \times 1.193 \times (1.193 + 2.79)/2 - W_1 \times 2.79^2/8$$

$$= 107.36 \text{ kNm}$$

- $R = M_d / bd^2$, where $b = 1000 \text{ mm}$, $d = 269 \text{ mm}$
 $= 1.4836$

$$P_t = A_{st} \times 100 / bd = \frac{20}{2 \times 415} \left(1 - \left(1 - 4.598 \times \frac{1.4836}{20} \right)^{1/2} \right)$$

$$= 593.71 \text{ mm}^2$$

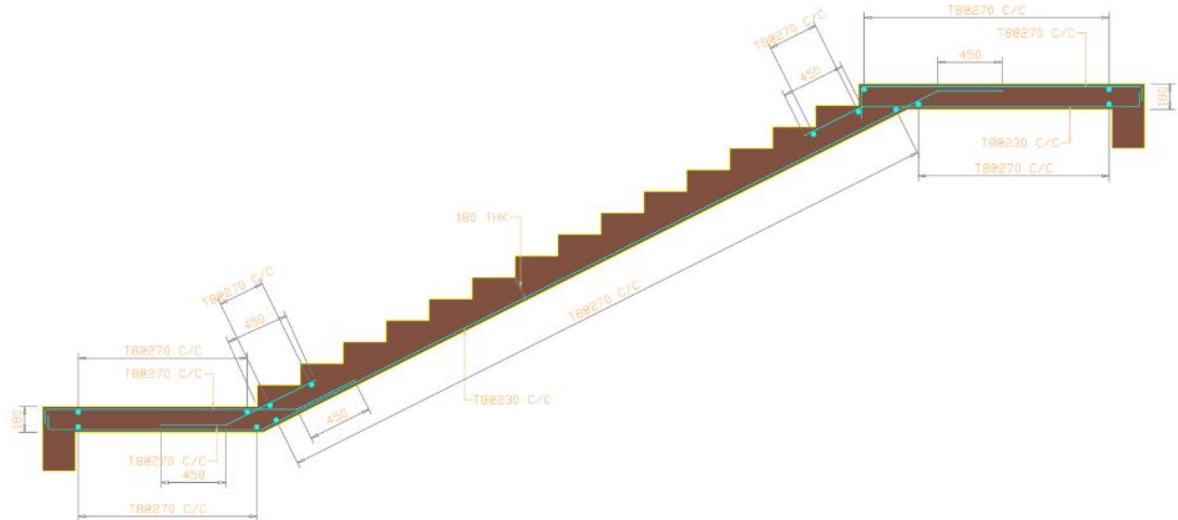
$$\text{Required spacing} = 1000 \times 113 / 593.71 = 190.32 \text{ mm}$$

- Spacing requirements according to IS 456 cl. 26.3.3 :
Spacing < 3 deff or 300mm
> main bar diameter
> max aggregate size + 5mm

Hence, ok

- Main reinforcement
Provide 12 dia. bar @ 191 mm spacing i.e. 5 bars

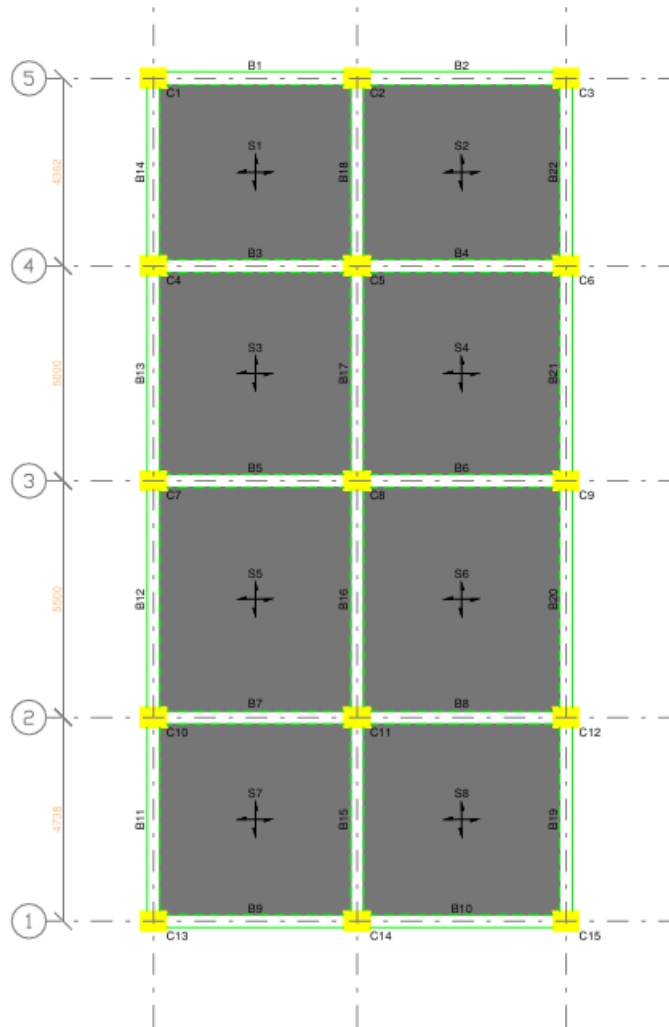
Reinforcement Detailing



SLAB DESIGN:

- Grade = M30, Fe 500
- Diameter of Reinforcement = 12 mm
- Clear Cover = 25 mm
- Thickness = 180 mm
- Supporting Beam Width of Longer/shorter span panel = 400 mm
- Imposed load = 1.5
- Live load = 2
- Two way slab design is carried out which is considered safe if actual span to eff. Depth ratio is lesser than permissible span to eff. Depth ratio.

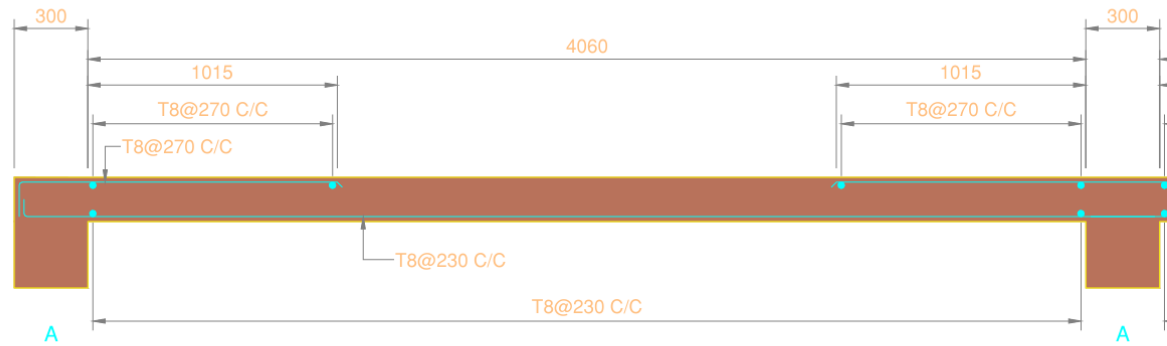
Slab layout:



PLAN AT 3.3 M

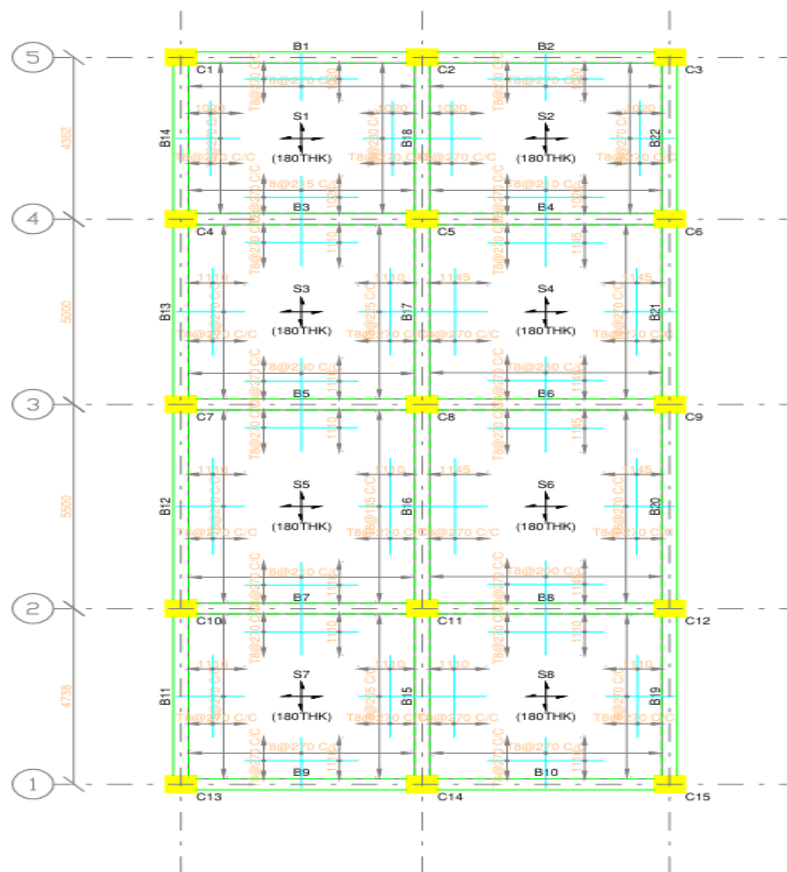
No	Slab	Thickness (mm)	Conc Grade	Steel Grade	Bottom @ Lx	Bottom @ Ly	Top @ Lx (Cont)	Top @ Lx (End)
1	S1	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 215	T8 @ 270
2	S2	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 210	T8 @ 270
3	S3	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 225	T8 @ 270
4	S4	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 225	T8 @ 270
5	S5	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 190	T8 @ 270
6	S6	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 185	T8 @ 270
7	S7	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 210	T8 @ 270
8	S8	180	M25	Fe415	T8 @ 230	T8 @ 230	T8 @ 200	T8 @ 270

Design:



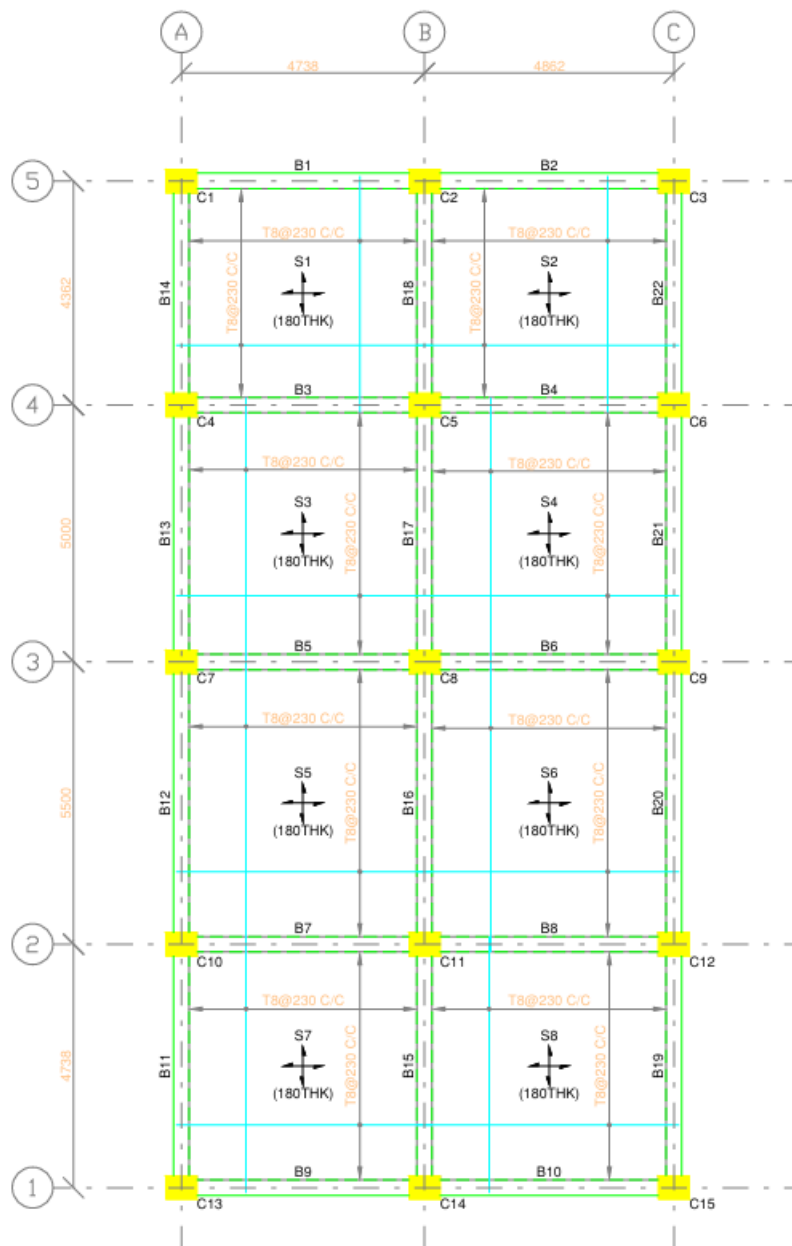
SLAB S1
(TWO WAY) (180 THK)

Top Reinforcement layout:



TOP REINFORCEMENT LAYOUT - 3.3 M
(SCALE: H = 1:100 / V = 1:100)

Bottom Reinforcement layout:



Command File

STAAD SPACE DXF IMPORT OF 8 MARCH SDD AUTOCAD

START JOB INFORMATION

ENGINEER DATE 08-Mar-26

END JOB INFORMATION

INPUT WIDTH 79

UNIT METER KN

JOINT COORDINATES

1 0 0 0; 2 4.73755 0 0; 3 0 0 4.36243; 4 4.73755 0 4.36243; 5 9.6001 0 0;
6 9.6001 0 4.36243; 7 0 0 9.36243; 8 4.73755 0 9.36243; 9 9.6001 0 9.36243;
10 0 0 14.8624; 11 4.73755 0 14.8624; 12 9.6001 0 14.8624; 13 0 0 19.6;
14 4.73755 0 19.6; 15 9.6001 0 19.6; 16 0 3.3 0; 17 4.73755 3.3 0;
18 0 3.3 4.36243; 19 4.73755 3.3 4.36243; 20 9.6001 3.3 0;
21 9.6001 3.3 4.36243; 22 0 3.3 9.36243; 23 4.73755 3.3 9.36243;
24 9.6001 3.3 9.36243; 25 0 3.3 14.8624; 26 4.73755 3.3 14.8624;
27 9.6001 3.3 14.8624; 28 0 3.3 19.6; 29 4.73755 3.3 19.6; 30 9.6001 3.3 19.6;
31 0 6.6 0; 32 4.73755 6.6 0; 33 0 6.6 4.36243; 34 4.73755 6.6 4.36243;
35 9.6001 6.6 0; 36 9.6001 6.6 4.36243; 37 0 6.6 9.36243;
38 4.73755 6.6 9.36243; 39 9.6001 6.6 9.36243; 40 0 6.6 14.8624;
41 4.73755 6.6 14.8624; 42 9.6001 6.6 14.8624; 43 0 6.6 19.6;
44 4.73755 6.6 19.6; 45 9.6001 6.6 19.6; 46 0 9.9 0; 47 4.73755 9.9 0;
48 0 9.9 4.36243; 49 4.73755 9.9 4.36243; 50 9.6001 9.9 0;
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57 9.6001 9.9 14.8624; 58 0 9.9 19.6; 59 4.73755 9.9 19.6; 60 9.6001 9.9 19.6;
61 0 13.2 0; 62 4.73755 13.2 0; 63 0 13.2 4.36243; 64 4.73755 13.2 4.36243;
65 9.6001 13.2 0; 66 9.6001 13.2 4.36243; 67 0 13.2 9.36243;
68 4.73755 13.2 9.36243; 69 9.6001 13.2 9.36243; 70 0 13.2 14.8624;
71 4.73755 13.2 14.8624; 72 9.6001 13.2 14.8624; 73 0 13.2 19.6;
74 4.73755 13.2 19.6; 75 9.6001 13.2 19.6; 76 0 16.5 0; 77 4.73755 16.5 0;
78 0 16.5 4.36243; 79 4.73755 16.5 4.36243; 80 9.6001 16.5 0;
81 9.6001 16.5 4.36243; 82 0 16.5 9.36243; 83 4.73755 16.5 9.36243;
84 9.6001 16.5 9.36243; 85 0 16.5 14.8624; 86 4.73755 16.5 14.8624;

87 9.6001 16.5 14.8624; 88 0 16.5 19.6; 89 4.73755 16.5 19.6;
 90 9.6001 16.5 19.6; 91 0 -3 0; 92 4.73755 -3 0; 93 0 -3 4.36243;
 94 4.73755 -3 4.36243; 95 9.6001 -3 0; 96 9.6001 -3 4.36243; 97 0 -3 9.36243;
 98 4.73755 -3 9.36243; 99 9.6001 -3 9.36243; 100 0 -3 14.8624;
 101 4.73755 -3 14.8624; 102 9.6001 -3 14.8624; 103 0 -3 19.6;
 104 4.73755 -3 19.6; 105 9.6001 -3 19.6;

MEMBER INCIDENCES

1 1 2; 2 1 3; 3 3 4; 4 2 4; 5 2 5; 6 5 6; 7 4 6; 8 3 7; 9 7 8; 10 4 8; 11 8 9;
 12 6 9; 13 7 10; 14 10 11; 15 11 8; 16 11 12; 17 9 12; 18 10 13; 19 13 14;
 20 11 14; 21 14 15; 22 12 15; 23 1 16; 24 2 17; 25 3 18; 26 4 19; 27 5 20;
 28 6 21; 29 7 22; 30 8 23; 31 9 24; 32 10 25; 33 11 26; 34 12 27; 35 13 28;
 36 14 29; 37 15 30; 38 16 17; 39 16 18; 40 18 19; 41 17 19; 42 17 20; 43 20 21;
 44 19 21; 45 18 22; 46 22 23; 47 19 23; 48 23 24; 49 21 24; 50 22 25; 51 25 26;
 52 26 23; 53 26 27; 54 24 27; 55 25 28; 56 28 29; 57 26 29; 58 29 30; 59 27 30;
 60 16 31; 61 17 32; 62 18 33; 63 19 34; 64 20 35; 65 21 36; 66 22 37; 67 23 38;
 68 24 39; 69 25 40; 70 26 41; 71 27 42; 72 28 43; 73 29 44; 74 30 45; 75 31 32;
 76 31 33; 77 33 34; 78 32 34; 79 32 35; 80 35 36; 81 34 36; 82 33 37; 83 37 38;
 84 34 38; 85 38 39; 86 36 39; 87 37 40; 88 40 41; 89 41 38; 90 41 42; 91 39 42;
 92 40 43; 93 43 44; 94 41 44; 95 44 45; 96 42 45; 97 31 46; 98 32 47; 99 33 48;
 100 34 49; 101 35 50; 102 36 51; 103 37 52; 104 38 53; 105 39 54; 106 40 55;
 107 41 56; 108 42 57; 109 43 58; 110 44 59; 111 45 60; 112 46 47; 113 46 48;
 114 48 49; 115 47 49; 116 47 50; 117 50 51; 118 49 51; 119 48 52; 120 52 53;
 121 49 53; 122 53 54; 123 51 54; 124 52 55; 125 55 56; 126 56 53; 127 56 57;
 128 54 57; 129 55 58; 130 58 59; 131 56 59; 132 59 60; 133 57 60; 134 46 61;
 135 47 62; 136 48 63; 137 49 64; 138 50 65; 139 51 66; 140 52 67; 141 53 68;
 142 54 69; 143 55 70; 144 56 71; 145 57 72; 146 58 73; 147 59 74; 148 60 75;

149 61 62; 150 61 63; 151 63 64; 152 62 64; 153 62 65; 154 65 66; 155 64 66;
156 63 67; 157 67 68; 158 64 68; 159 68 69; 160 66 69; 161 67 70; 162 70 71;
163 71 68; 164 71 72; 165 69 72; 166 70 73; 167 73 74; 168 71 74; 169 74 75;
170 72 75; 171 61 76; 172 62 77; 173 63 78; 174 64 79; 175 65 80; 176 66 81;
177 67 82; 178 68 83; 179 69 84; 180 70 85; 181 71 86; 182 72 87; 183 73 88;
184 74 89; 185 75 90; 186 76 77; 187 76 78; 188 78 79; 189 77 79; 190 77 80;
191 80 81; 192 79 81; 193 78 82; 194 82 83; 195 79 83; 196 83 84; 197 81 84;
198 82 85; 199 85 86; 200 86 83; 201 86 87; 202 84 87; 203 85 88; 204 88 89;
205 86 89; 206 89 90; 207 87 90; 208 1 91; 209 2 92; 210 3 93; 211 4 94;
212 5 95; 213 6 96; 214 7 97; 215 8 98; 216 9 99; 217 10 100; 218 11 101;
219 12 102; 220 13 103; 221 14 104; 222 15 105;

DEFINE MATERIAL START

ISOTROPIC CONCRETE

E 2.17185e+07

POISSON 0.17

DENSITY 23.5616

ALPHA 1e-05

DAMP 0.05

G 9.28139e+06

TYPE CONCRETE

STRENGTH FCU 27579

END DEFINE MATERIAL

MEMBER PROPERTY

1 TO 22 38 TO 59 PRIS YD 0.6 ZD 0.3

75 TO 96 112 TO 133 PRIS YD 0.45 ZD 0.3

149 TO 170 186 TO 207 PRIS YD 0.3 ZD 0.23

23 TO 37 60 TO 74 97 TO 111 134 TO 148 171 TO 185 208 TO 221 -
222 PRIS YD 0.6 ZD 0.45
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
91 TO 105 FIXED
DEFINE WIND LOAD
TYPE 1 WIND 1
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
CUSTOM:PARAMS
!> END GENERATED DATA BLOCK
INT 1.653 1.653 1.811 HEIG 0 10 16.5
EXP 1 JOINT 1 TO 105
DEFINE IS1893 2016 LOAD
ZONE 0.16 RF 5 I 1 ST 1 PX 0.5 PZ 0.5 DT 3 SA 2.5 DF 1 HT 16.5 DX 9.6 DZ 19.6
FLOOR WEIGHT
YRANGE 0 16 FLOAD 6.75 XRANGE 0 9.6 ZRANGE 0 19.6
LOAD 1 LOADTYPE Seismic-H TITLE EQ +X
1893 LOAD X 1
LOAD 2 LOADTYPE Seismic-H TITLE EQ -X
1893 LOAD X -1
LOAD 3 LOADTYPE Seismic-H TITLE EQ +Z
1893 LOAD Z 1
LOAD 4 LOADTYPE Seismic-H TITLE EQ -Z
1893 LOAD Z -1
LOAD 5 LOADTYPE Dead TITLE Dead load

MEMBER LOAD

24 26 30 33 36 38 40 TO 42 44 46 TO 48 51 TO 53 56 TO 58 61 63 67 70 73 75 -
77 TO 79 81 83 TO 85 88 TO 90 93 TO 95 98 100 104 107 110 112 114 TO 116 -
118 120 TO 122 125 TO 127 130 TO 132 135 137 141 144 147 149 151 TO 153
155 -

157 TO 159 162 TO 164 167 TO 169 172 174 178 181 184 UNI GY -9.9

1 3 TO 5 7 9 TO 11 14 TO 16 19 TO 21 23 25 27 TO 29 31 32 34 35 37 39 43 45 -
49 50 54 55 59 60 62 64 TO 66 68 69 71 72 74 76 80 82 86 87 91 92 96 97 99 -
101 TO 103 105 106 108 109 111 113 117 119 123 124 128 129 133 134 136
138 -

139 TO 140 142 143 145 146 148 150 154 156 160 161 165 166 170 171 173
175 -

176 TO 177 179 180 182 183 185 TO 207 UNI GY -15.18

FLOOR LOAD

YRANGE 0 13.2 FLOAD -6 X RANGE 0 9.6 Z RANGE 0 19.6 GY

YRANGE 14 16.5 FLOAD -6 X RANGE 0 9.6 Z RANGE 0 19.6 GY

SELFWEIGHT Y -1 LIST ALL

LOAD 6 LOADTYPE Live TITLE Live load

FLOOR LOAD

YRANGE 0 16.5 FLOAD -3 X RANGE 0 9.6 Z RANGE 0 19.6 GY

LOAD 7 LOADTYPE Wind TITLE Wind load

WIND LOAD X 1.4 TYPE 1 XR 0 9.6 YR 0 16.5 ZR 0 19.6

LOAD COMB 8 COMB - 1.5 Dead + 1.5 Live

5 1.5 6 1.5

LOAD COMB 9 COMB - 1.2 Dead + 1.2 Live + 1.2 Wind

5 1.2 6 1.2 7 1.2

LOAD COMB 10 COMB - 1.2 Dead + 1.2 Live + -1.2 Wind

5 1.2 6 1.2 7 -1.2

LOAD COMB 11 COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (1)

5 1.2 6 1.2 1 1.2

LOAD COMB 12 COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (2)

5 1.2 6 1.2 2 1.2

LOAD COMB 13 COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (3)

5 1.2 6 1.2 3 1.2

LOAD COMB 14 COMB - 1.2 Dead + 1.2 Live + 1.2 Seismic-H (4)

5 1.2 6 1.2 4 1.2

LOAD COMB 15 COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (1)

5 1.2 6 1.2 1 -1.2

LOAD COMB 16 COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (2)

5 1.2 6 1.2 2 -1.2

LOAD COMB 17 COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (3)

5 1.2 6 1.2 3 -1.2

LOAD COMB 18 COMB - 1.2 Dead + 1.2 Live + -1.2 Seismic-H (4)

5 1.2 6 1.2 4 -1.2

LOAD COMB 19 COMB - 1.5 Dead + 1.5 Wind

5 1.5 7 1.5

LOAD COMB 20 COMB - 1.5 Dead + -1.5 Wind

5 1.5 7 -1.5

LOAD COMB 21 COMB - 1.5 Dead + 1.5 Seismic-H (1)

5 1.5 1 1.5

LOAD COMB 22 COMB - 1.5 Dead + 1.5 Seismic-H (2)

5 1.5 2 1.5

LOAD COMB 23 COMB - 1.5 Dead + 1.5 Seismic-H (3)

5 1.5 3 1.5

LOAD COMB 24 COMB - 1.5 Dead + 1.5 Seismic-H (4)

5 1.5 4 1.5

LOAD COMB 25 COMB - 1.5 Dead + -1.5 Seismic-H (1)

5 1.5 1 -1.5

LOAD COMB 26 COMB - 1.5 Dead + -1.5 Seismic-H (2)

5 1.5 2 -1.5

LOAD COMB 27 COMB - 1.5 Dead + -1.5 Seismic-H (3)

5 1.5 3 -1.5

LOAD COMB 28 COMB - 1.5 Dead + -1.5 Seismic-H (4)

5 1.5 4 -1.5

LOAD COMB 29 COMB - 0.9 Dead + 1.5 Wind

5 0.9 7 1.5

LOAD COMB 30 COMB - 0.9 Dead + -1.5 Wind

5 0.9 7 -1.5

LOAD COMB 31 COMB - 0.9 Dead + 1.5 Seismic-H (1)

5 0.9 1 1.5

LOAD COMB 32 COMB - 0.9 Dead + 1.5 Seismic-H (2)

5 0.9 2 1.5

LOAD COMB 33 COMB - 0.9 Dead + 1.5 Seismic-H (3)

5 0.9 3 1.5

LOAD COMB 34 COMB - 0.9 Dead + 1.5 Seismic-H (4)

5 0.9 4 1.5

LOAD COMB 35 COMB - 0.9 Dead + -1.5 Seismic-H (1)

5 0.9 1 -1.5

LOAD COMB 36 COMB - 0.9 Dead + -1.5 Seismic-H (2)

5 0.9 2 -1.5

LOAD COMB 37 COMB - 0.9 Dead + -1.5 Seismic-H (3)

5 0.9 3 -1.5

LOAD COMB 38 COMB - 0.9 Dead + -1.5 Seismic-H (4)

5 0.9 4 -1.5

PERFORM ANALYSIS PRINT ALL

PERFORM ANALYSIS PRINT ALL

PERFORM ANALYSIS PRINT ALL

PERFORM ANALYSIS PRINT ALL

START CONCRETE DESIGN

CODE INDIAN

DESIGN BEAM 1 3 5 7 9 11 14 16 19 21 38 40 42 44 46 48 51 53 56 58 75 77 79

-

81 83 85 88 90 93 95 112 114 116 118 120 122 125 127 130 132 149 151 TO
153 -

155 157 TO 159 162 TO 164 167 TO 169 186 188 TO 190 192 194 TO 196 -

199 TO 201 204 TO 206

DESIGN COLUMN 2 4 6 8 10 12 13 15 17 18 20 22 TO 37 39 41 43 45 47 49 50
52 -

54 55 57 59 TO 74 76 78 80 82 84 86 87 89 91 92 94 96 TO 111 113 115 117 -

119 121 123 124 126 128 129 131 133 TO 148 150 152 154 156 158 160 161
163 -

165 166 168 170 TO 185 208 TO 222

DESIGN COLUMN 2 4 6 8 10 12 13 15 17 18 20 22 TO 37 39 41 43 45 47 49 50
52 -

54 55 57 59 TO 74 76 78 80 82 84 86 87 89 91 92 94 96 TO 111 113 115 117 -

119 121 123 124 126 128 129 131 133 TO 148 150 152 154 156 158 160 161
163 -

165 166 168 170 TO 185 208 TO 222

DESIGN BEAM 1 TO 22 38 TO 59 75 TO 96 112 TO 133 149 TO 170 186 TO 207

DESIGN COLUMN 2 4 6 8 10 12 13 15 17 18 20 22 TO 37 39 41 43 45 47 49 50
52 -

54 55 57 59 TO 74 76 78 80 82 84 86 87 89 91 92 94 96 TO 111 113 115 117 -

119 121 123 124 126 128 129 131 133 TO 148 150 152 154 156 158 160 161
163 -

165 166 168 170 TO 185 187 193 198 203 208 TO 222

END CONCRETE DESIGN

FINISH

